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# Determining the charge of an electron with oil drops

Carissa Kiehl
*Department of Physics and Astronomy, Carthage College, 2001 Alford Park Drive, Kenosha, WI 53140, USA*

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This experiment determines the charge of an electron by measuring the velocities of ionized oil drops in a gravitational and electric field. Through these measurements and an analysis of the forces at play, the charge of the oil drop can be calculated. While each oil drop may have a different number of excess electrons, they will all be integer multiples of the charge of a single electron. Therefore, we can take the greatest common multiple of all calculated charges to be the charge of a single electron.

## I. INTRODUCTION

Ever since the Greeks first discovered the effects of electricity, the electron became the focus of much scientific research. J.J. Thompson proved the existence of electrons as unique particles in his cathode ray experiment [1]. In the same experiment, Thompson came to an estimation of the electron charge-to-mass ratio. He concluded the electron has a charge of about $6.8\times10^{-10}$ esu. Thompson's further research also helped establish the plum pudding model, which accounted for electrons in the model of an atom [2]. While Thompson's contributions are significant, our experiment will build on his results and establish a more accurate determination of the electron charge.

Our experiment involves analyzing the behavior of small oil droplets in a gravitational and electric field. Because oil drops quickly reach terminal velocity in these fields, we can equate the forces at play. In the case of an oil drop falling in a gravitational field, the drag force and gravitational force are equal and we arrive at the following equation:

$$mg = kv_f \tag{1}$$

where $m$ is the mass of the oil drop, $g$ is the acceleration due to gravity, $k$ is the drag coefficient, and $v_f$ is the falling velocity of the drop. In the case of an electric field, the electrical force would be equal to the gravitational force and drag force and we arrive at the following equation:

$$Eq = mg + kv_r \tag{2}$$

where $E$ is the electric intensity, $q$ is the charge of the oil drop, and $v_r$ is the rising velocity of the drop. Putting these equations together and solving for $q$ yields the following equation:

$$q = \frac{mg(v_f + v_r)}{Ev_f}. \tag{3}$$

However, we need to eliminate $m$ from the equation since that quantity is not easily measurable for low-mass oil drops. To do this, we can write mass in terms of a sphere's density times its volume. This calculation requires the radius of the oil drop which we can determine by Stokes' Law. Stokes' Law relates the velocity of a particle falling in a viscous medium to the particle's radius. While the derivation using Stokes' Law gets quite convoluted, the resulting equation for $q$ simplifies to the following:

$$q = \frac{\frac{4}{3}dg\pi\rho\left(v_f + v_r\right)\left(-\frac{b}{2p} + \sqrt{\frac{b^2}{4p^2} + \frac{9\eta v_f}{2g\rho}}\right)^3}{Vv_f} \tag{4}$$

where $d$ is height of the chamber apparatus, $\rho$ is the density of the oil, $\eta$ is the viscosity of air, $b$ is a viscosity constant, $p$ is the pressure, and $V$ is the voltage. All variables are in SI units, and the resulting charge $q$ is in Coulombs.

The experiment apparatus includes a chamber for the oil drops that induces an electric field. The chamber can also introduce an ionization source that changes the number of excess electrons on the oil drops. The rising and falling velocities of oil drops can be measured through a viewing scope with the electric field on and off respectively. In the end, Equation 4 is used to calculate the charge of each oil drop. While the charges may vary, the greatest common multiple of all calculated charges is the charge of a single electron.

## II. CALCULATED CHARGES

The calculated charges of the oil drops appeared to be separated into several distinct clusters based on their values. We took the cluster made up of the smallest values, determined the average, and took the average to be our temporary charge of a single electron. Because the charge of an oil drop can only be an integer multiple of a single electron charge, we used our temporary value to predict the number of electrons in the remaining calculated charges. Finally, to get an accurate value for the charge of a single electron, we plotted the predicted number of electrons in the oil drops against their calculated

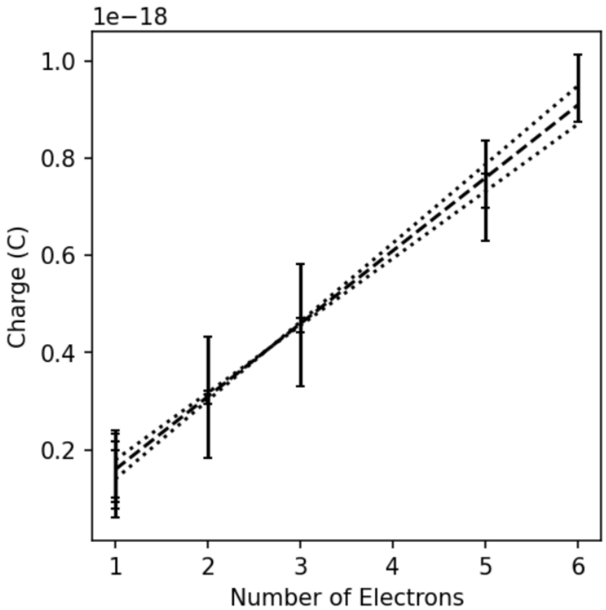


FIG. 1. The calculated charge for each oil drop is plotted against its predicted number of excess electrons. The slope of the trendline is $1.5 \times 10^{-19}$ C/e and represents the charge of a single electron. The dashed line represents the uncertainty in the slope which is $\pm\ 1 \times 10^{-20}$ C/e. The uncertainty in charge for each oil drop is shown by error bars.

charges. The resulting trendline is shown in Figure 1. Because the slope of the least squares regression trendline represents the increase in charge of an oil drop as one more electron is added, it can be interpreted as the charge of a single electron. The slope of the trendline, and therefore the value for the elementary charge, was determined to be

$$q_e = 1.5 \times 10^{-19} \pm 1 \times 10^{-20}\ \mathrm{C}$$

. The other calculated charges are integer multiples of this charge. While there is some variance in the calculated charges from the expected value, there is still a strong linear correlation between all values, further proving the calculated value of the elementary charge.

Our calculated results were fairly accurate, but not as precise. This can be attributed to the measurement of the rising and falling velocities of the oil drops. There was human error involved in these measurements as they depended on reaction time. However, because we ran many trials, the random error became less significant and the varying results still generated a well-correlated trendline that accurately predicts the charge of an electron.

## III. CONCLUSION

Through this experiment, we were able to calculate the charge of individual oil drops by measuring their falling velocities in a gravitational field and rising velocities in an electric field. By grouping similar calculated charges and predicting how many excess electrons each oil drop had, we were able to fit a trendline to the number of electrons in the oil drops and their respective charges. The slope of the trendline represents the charge of a single electron and was determined to be $1.5 \times 10^{-19} \pm 1 \times 10^{-20}$ C.

The implications of our results are significant. They establish that electric charge is quantized and that all calculated charges are only integer multiples of the elementary charge. The charge of an electron is also a fundamental constant that can be used to calculate many other physics constants. This research could be expanded to increase the precision of the determined charge value. Also, new methods can be researched to find how to measure the charge of other fundamental subatomic particles.

### Appendix A: Supplemental Information

The uncertainty in the calculated charges was estimated based on the uncertainties in measuring the rising and falling velocities of the oil drops. While there were uncertainties in the given values for the chamber apparatus and the density of the oil used, these uncertainties were negligible compared to the uncertainties induced by human error in measuring the velocities with a stopwatch. Because we only consider uncertainty in the velocities, the equation to determine the overall uncertainty in the charge $\sigma_q$ is the following:

$$\sigma_q = \sqrt{\left(\frac{dq}{dv_r}\sigma_{v_r}\right)^2 + \left(\frac{dq}{dv_f}\sigma_{v_f}\right)^2} \tag{A1}$$

where $dq/dv_r$ is the derivative of Equation 4 with respect to the rising velocity and $dq/dv_f$ is the derivative of Equation 4 with respect to the falling velocity. We determined $\pm\ 7 \times 10^{-6}$ m/s would be an adequate uncertainty value for the measured velocities. This value is based on the range of velocities we measured for the same oil drops. We used least squares regression to determine the charge of an electron. Factoring in the uncertainty from the measured velocities, the uncertainty in the slope of the trendline, and therefore the charge of an electron, was determined to be $\pm\ 1 \times 10^{-20}$ C/e. As seen in Figure 1, the error bars for all the oil drops measured contain the expected value for charge. Therefore, our model accurately predicts the range in which the true charge of an electron resides.

[1] J. Thompson, On the structure of the atom: an investigation of the stability and periods of oscillation of a number of corpuscles arranged at equal intervals around the cir-

cumference of a circle; with application of the results to the theory of atomic structure (1904).
[2] J. Thompson, Cathode rays (1897).